

# Digital Forensics I Practical Slips

## Easy Answers - Slightly More Explained

This version keeps the answers easy and exam-friendly, but explains each point a little more than the previous PDF.

**Included sheets:** 1-6, 9, 11-20, 26, 27 and 29.

**Excluded as requested:** 7, 8, 10, 21, 22, 23, 24, 25, 28 and 30.

**Q3 Viva:** Not included.

## Sheet No. 1

**Q1. Install and configure Autopsy on the forensic workstation. Create a new case and add a sample data source. Prepare a report with screenshots.**

**Answer / Procedure:**

- Download Autopsy from the official source and install it on the forensic workstation.
- Open Autopsy and choose **New Case**.
- Enter the case name, case number and examiner details. Select a folder where the case will be stored.
- After the case is created, click **Add Data Source**.
- Select the sample source, such as a disk image, local disk or logical files, and browse to its location.
- Select suitable ingest modules. These modules automatically search for items such as deleted files, web activity and file metadata.
- Start the analysis and wait until processing is complete.
- Explore **Data Sources**, **Views** and **Results** to check files, folders and artifacts.
- Take screenshots of the main steps and important findings for the practical report.
- **Result:** Autopsy was installed, a forensic case was created and the sample data source was added successfully for analysis.

**Q2. Explain the concept of digital evidence and identify at least five types of digital evidence found in a computer system.**

**Answer:**

- **Digital evidence** is any information stored, processed or transmitted in digital form that can help prove or explain an event during an investigation.
- It must be collected carefully because digital data can be changed, copied or deleted easily.
- Common types include **documents** such as PDF/DOCX files, **images and videos**, **emails**, **browser history**, **system and application logs**, **chat messages**, **deleted files** and **file metadata**.
- Investigators preserve this evidence and verify it using methods such as hashing.

## Sheet No. 2

**Q1. Install and configure FTK Imager. Explore its interface and identify the options used for evidence acquisition. Prepare a report.**

**Answer / Procedure:**

- Download and install FTK Imager on the forensic workstation.
- Open FTK Imager with administrator permission.
- Study the main interface. Important areas include **Evidence Tree**, **File List**, **Viewer** and **Properties**.
- To acquire evidence, go to **File > Create Disk Image**.
- Choose the correct source type, for example Physical Drive, Logical Drive or an existing Image File.
- Select the correct device and then choose an image format such as **E01** or **RAW (dd)**.
- Enter case information, examiner name and destination folder.
- Enable **Verify images after they are created** so the copy can be checked after acquisition.
- Start the imaging process and record the hash values and completion details.
- **Result:** FTK Imager was installed and its main evidence-acquisition options were explored successfully.

**Q2. Explain the importance of a controlled digital forensic environment during investigations.**

**Answer:**

- A controlled forensic environment is important because evidence must not be changed accidentally during examination.
- Only authorized investigators should have access to the evidence and forensic workstation.
- Trusted forensic tools and write blockers help prevent modification of the original storage device.
- All activities should be documented so another examiner can understand what was done.
- A controlled environment protects evidence integrity, supports chain of custody and makes the investigation more reliable.

## Sheet No. 3

**Q1. Create a forensic case in Autopsy and add a disk image. Identify files, folders, and basic artifacts. Document the findings.**

**Answer / Procedure:**

- Open Autopsy and select **New Case**.
- Enter the case name, case number, examiner details and storage location.
- Choose **Add Data Source > Disk Image or VM File**.
- Browse to the provided disk image and add it to the case.
- Select suitable ingest modules and start processing.
- After processing, open **Data Sources** to view the folder structure and normal files.
- Check **Views** and **Results** for basic artifacts such as deleted files, recent documents, web history, images and metadata.
- Record important file names, paths, sizes and timestamps.
- Add screenshots of the case creation, added image and important findings.
- **Result:** The disk image was added to Autopsy and basic files, folders and forensic artifacts were identified.

**Q2. Explain the role of a forensic investigator in preserving digital evidence.**

**Answer:**

- A forensic investigator must protect digital evidence from alteration, damage or unauthorized access.
- The investigator normally works on a forensic copy instead of directly examining the original device.
- Hash values are calculated before and after acquisition to check whether the evidence has changed.
- Every collection, transfer and examination step should be documented in the chain-of-custody record.
- Evidence should be stored securely, labelled correctly and accessed only by authorized people.

## Sheet No. 4

**Q1. Explore the digital forensic environment and identify available tools for disk analysis, file recovery, and network analysis. Prepare a report.**

**Answer / Procedure:**

- Open the forensic workstation and check the forensic tools installed in the environment.
- For **disk analysis**, identify tools such as Autopsy and FTK Imager. These tools help examine disk images, files, partitions and metadata.
- For **file recovery**, use Autopsy recovery features or other available recovery tools to find deleted data.
- For **network analysis**, identify Wireshark, which captures and examines network packets.
- Open each tool and note its main interface and important functions.
- Write a short table in the report showing the tool name, category and purpose.
- **Result:** Tools for disk analysis, deleted-file recovery and network investigation were identified in the forensic environment.

**Q2. Explain the need and importance of digital forensics in cyber security.**

**Answer:**

- Digital forensics helps investigate cyber attacks, data theft, malware incidents and unauthorized access.
- It helps investigators determine **what happened, when it happened, how it happened and which systems were affected**.
- It can recover deleted or hidden data and examine logs, browser history, files and network activity.
- The findings can support incident response, legal action and internal investigations.
- Forensic results also help organizations understand weaknesses and improve future security controls.

## Sheet No. 5

**Q1. Identify and classify digital evidence from a provided sample system (documents, images, logs, browser history, emails). Prepare a table of findings.**

**Answer / Procedure:**

| Evidence Type   | What to Check                                | Possible Evidential Value                  |
|-----------------|----------------------------------------------|--------------------------------------------|
| Documents       | PDF, DOCX, TXT, spreadsheets                 | User-created information, plans or records |
| Images          | JPG, PNG, screenshots                        | Visual proof and image metadata            |
| Logs            | System, security or application logs         | Shows system and application activity      |
| Browser History | Visited URLs, searches, downloads            | Shows internet activity and user interest  |
| Emails          | Sender, receiver, subject, time, attachments | Shows communication and possible intent    |

Open the sample system in the forensic tool and classify each useful item under the correct evidence type. Record file name, path, timestamp and any important observation.

**Result:** The available digital evidence was identified and classified in a structured table.

**Q2. Explain the characteristics of digital evidence.**

**Answer:**

- Digital evidence can be copied exactly without losing quality.
- It is fragile because it can be modified, deleted or overwritten very quickly.
- It often contains useful **metadata** such as timestamps, file paths and owner information.
- Some evidence may be hidden, encrypted, compressed or deleted.
- Hash values can be used to show whether a digital file or forensic image has remained unchanged.
- Good documentation is required to prove where the evidence came from and how it was handled.

## Sheet No. 6

**Q1. Create a disk image of a removable drive using FTK Imager. Document each step with screenshots.**

**Answer / Procedure:**

- Connect the removable drive to the forensic workstation.
- Open FTK Imager and select **File > Create Disk Image**.
- Choose **Physical Drive** so that the whole removable device can be copied.
- Select the correct removable drive carefully to avoid imaging the wrong device.
- Choose the image format, such as **E01** or **RAW**.
- Enter the case information and select a destination folder with enough free space.
- Enable image verification and start the acquisition.
- After completion, record the image file name, size, MD5/SHA hash values and verification status.
- Take screenshots of source selection, image settings and final verification.
- **Result:** A forensic image of the removable drive was created and verified using FTK Imager.

**Q2. Explain the purpose of disk imaging in digital forensics.**

**Answer:**

- Disk imaging creates a **bit-by-bit copy** of a storage device.
- This allows the investigator to work on a copy while keeping the original evidence safe and unchanged.
- A forensic image may include active files, deleted data, file-system information and unallocated space.
- Hash values are used to verify that the image matches the original source.
- Disk imaging is important because it preserves evidence for repeated analysis and future review.

## Sheet No. 9

**Q1. Generate MD5 and SHA-256 hash values for the original media and the acquired image. Verify integrity and document the results.**

**Answer / Procedure:**

- Open the original evidence media using a trusted forensic tool such as FTK Imager.
- Generate or note the **MD5** and **SHA-256** values of the original media.
- Open the acquired forensic image and calculate the same two hash values.
- Write the source hash values and image hash values in a comparison table.
- Compare MD5 of the source with MD5 of the image and SHA-256 of the source with SHA-256 of the image.
- If the corresponding values are identical, the image is considered an exact and unchanged copy of the source.
- If they do not match, the acquisition must be checked because the evidence may have changed or the copy may be incomplete.
- **Result:** Matching MD5 and SHA-256 values confirm the integrity of the forensic acquisition.

**Q2. Explain the importance of hash values in forensic investigations.**

**Answer:**

- A hash value is a digital fingerprint calculated from data.
- Even a very small change in the data normally produces a different hash value.
- Investigators calculate hashes before and after copying evidence to prove that the forensic copy matches the original.
- Hash values also help detect accidental or intentional modification during storage or analysis.
- Common algorithms used in forensics include MD5 and the stronger SHA-256 algorithm.

## Sheet No. 11

**Q1. Analyze a FAT file system and identify directory entries, file attributes, and deleted file indicators.**

**Answer / Procedure:**

- Open the FAT disk image in Autopsy or another forensic tool.
- Browse the directory tree and identify files and folders.
- Inspect directory entries for information such as file name, file size, starting cluster and timestamps.
- Check file attributes such as **Read-only**, **Hidden**, **System**, **Directory** and **Archive**.
- Examine deleted entries. In FAT, the first character/byte of a deleted directory entry is normally marked as deleted.
- Check whether deleted file data still exists in clusters that have not yet been overwritten.
- Record important normal, hidden and deleted entries in the report.
- **Result:** FAT directory entries, attributes and deleted-file indicators were identified.

**Q2. Explain the structure of the FAT file system.**

**Answer:**

- FAT stands for **File Allocation Table** and is used to manage files on a storage device.
- Its main parts include the **Boot Sector**, one or more **FAT tables**, directory information and the **data area**.
- The FAT table keeps track of which clusters are free, used or linked to a file.
- Directory entries store information such as file name, file size, attributes, timestamps and starting cluster.
- The data area contains the actual contents of files and folders.

## Sheet No. 12

### Q1. Analyze an NTFS file system and identify MFT entries, timestamps, and file permissions.

#### Answer / Procedure:

- Open the NTFS forensic image in Autopsy.
- Browse the data source and select important files or folders.
- Inspect the file metadata and MFT-related information shown by the tool.
- The **Master File Table (MFT)** contains a record for almost every file and folder on an NTFS volume.
- Record available timestamps such as Created, Modified and Accessed. Some tools may also show metadata-change information.
- Check file owner and permissions where this information is available.
- Look for deleted, unusual or suspicious entries and record their paths and metadata.
- **Result:** NTFS MFT information, timestamps and file permissions were examined and documented.

### Q2. Explain the structure of the NTFS file system.

#### Answer:

- NTFS is a Windows file system designed to support security, large files and reliable storage.
- Its most important structure is the **Master File Table (MFT)**, which stores records for files and folders.
- Each record can contain file names, timestamps, security information, data location and other attributes.
- NTFS supports permissions, ownership, journaling, compression and other advanced features.
- Other special metadata files help NTFS manage free space, logs and file-system information.

## Sheet No. 13

**Q1. Explore file metadata in Autopsy and identify created, modified, accessed, and changed timestamps. Prepare a report.**

**Answer / Procedure:**

- Open the forensic case in Autopsy and select the required data source.
- Choose an important file and open its **Metadata** or **Properties** view.
- Record the **Created** time, which usually shows when the file was created on that file system.
- Record the **Modified** time, which shows when file content was last changed.
- Record the **Accessed** time, which may show when the file was last opened or accessed.
- Record the **Changed** or metadata-change time if the file system/tool provides it.
- Repeat the same process for other relevant files and present the timestamps in a small table.
- **Result:** File metadata and important timestamps were identified and documented.

**Q2. Explain the importance of metadata in forensic investigations.**

**Answer:**

- Metadata means **data about data**. It describes a file even when the actual content is not opened.
- It may contain file name, size, location, owner, timestamps, device information or application details.
- Metadata helps investigators understand when files were created, modified or accessed.
- It can be used to build a timeline and connect a user or device with an activity.
- It can also reveal useful details that are not visible in the normal file content.

## Sheet No. 14

**Q1. Identify hidden and system files in a provided forensic image and document the findings.**

**Answer / Procedure:**

- Open the provided forensic image in Autopsy.
- Browse files and folders and check their file attributes and metadata.
- Look for files marked with the **Hidden** or **System** attribute.
- Check whether any hidden file is located in an unusual folder or has a misleading name or extension.
- Record the file name, full path, size, timestamps and available hash values.
- Preview suspicious files only through the forensic tool so the evidence is not changed.
- Document relevant findings and take screenshots if required.
- **Result:** Hidden and system files were identified and documented from the forensic image.

**Q2. Explain how hidden files may be used to conceal malicious or sensitive data.**

**Answer:**

- Attackers may use hidden files to make malware, scripts or stolen data less visible to a normal user.
- System or hidden attributes can stop files from appearing in normal folder views.
- Malicious files may also be placed in system folders or given names that look like legitimate Windows files.
- Sensitive data can be hidden in uncommon directories, archives or other less noticeable locations.
- Forensic tools show hidden/system entries and allow investigators to examine their metadata and content.

## Sheet No. 15

### Q1. Analyze file structure and storage allocation using Autopsy and identify file fragments or unusual file entries.

#### Answer / Procedure:

- Open the disk image in Autopsy and browse the file-system structure.
- Select files and inspect their size, metadata and storage information.
- Check whether deleted or unallocated data is present.
- Look for unusual file entries such as strange extensions, duplicate names, very large or very small files, and files in unexpected folders.
- Where the tool provides storage details, check whether file data is stored in one continuous area or different locations.
- Fragments may appear when parts of a file are stored in non-contiguous clusters.
- Record suspicious entries and explain why they are unusual.
- **Result:** File structure and storage allocation were examined for fragments, deleted data and unusual entries.

### Q2. Explain file allocation and storage mechanisms in file systems.

#### Answer:

- A storage device is divided into small storage units called blocks or clusters.
- The file system keeps records showing which clusters belong to each file.
- If enough continuous space is available, a file may be stored in nearby clusters.
- When free space is scattered, the file can be divided into parts; this is called **fragmentation**.
- When a file is deleted, its clusters may be marked free even though the old data can remain until overwritten.
- Different file systems use structures such as FAT tables or the NTFS MFT to manage this allocation.

## Sheet No. 16

**Q1. Recover deleted files using Autopsy and save the recovered evidence. Prepare a report with screenshots.**

**Answer / Procedure:**

- Open the forensic case and load the required disk image in Autopsy.
- Use **Deleted Files** or search/filter options to find files marked as deleted.
- Preview the file to check whether enough of its content is still available.
- Select the required deleted file and use the extraction/export option such as **Extract File(s)**.
- Save the recovered file to a separate evidence/output folder, not back to the source image.
- Record its original path, file name, size, timestamps and hash value if available.
- Take screenshots showing the deleted file in Autopsy and the recovery/export step.
- Do not change the original evidence while recovering the file.
- **Result:** Deleted files were located, recovered and saved separately for further examination.

**Q2. Explain deleted file recovery techniques in digital forensics.**

**Answer:**

- **File-system recovery** uses remaining file-system records to locate a deleted file.
- **File carving** searches raw disk data for known file signatures such as headers and footers.
- **Unallocated-space analysis** checks storage areas marked as free because deleted data may still remain there.
- Backups, restore points or shadow copies may contain older versions of files.
- Recovered files should be documented and hashed so their integrity can be checked later.

## Sheet No. 17

**Q1. Analyze recovered documents and identify their metadata, content, and relevance to the investigation.**

**Answer / Procedure:**

- Open each recovered document using a safe forensic viewer or analysis tool.
- Record the file name, original path, extension and file size.
- Check metadata such as author, created time, modified time and other available properties.
- Read the document content and note important names, dates, messages or other case-related information.
- Compare the document with other evidence, such as logs, emails or browser history, if available.
- Decide whether the document is highly relevant, possibly relevant or not relevant to the investigation.
- Record the observations without editing the recovered document.
- **Result:** Recovered documents were examined for metadata, content and their possible value to the investigation.

**Q2. Explain the importance of recovered artifacts during investigations.**

**Answer:**

- Recovered artifacts may contain information that a user deleted intentionally or accidentally.
- They can include documents, images, messages, browsing traces and other useful data.
- These artifacts help investigators reconstruct earlier activities on the device.
- They can support other evidence or show that another conclusion is incorrect.
- Even a small recovered item can provide a useful lead, timestamp, username or connection to another event.

## Sheet No. 18

**Q1. Recover deleted image files and verify their integrity using hash values. Prepare a report.**

**Answer / Procedure:**

- Open the forensic image in Autopsy and locate files marked as deleted.
- Filter or search for common image types such as JPG, JPEG and PNG.
- Preview the images to check whether they can still be recovered correctly.
- Extract the required deleted images to a separate evidence folder.
- Generate a hash value for each recovered image using the forensic tool or another trusted hashing utility.
- Record the file name, original path, file size, recovery status and hash value.
- If the same recovered copy is checked later and gives the same hash, it shows that the file has not changed.
- **Result:** Deleted images were recovered and their integrity was documented using hash values.

**Q2. Explain how investigators validate recovered digital evidence.**

**Answer:**

- Investigators document the exact source and method used to recover the evidence.
- They calculate hash values so later copies can be checked against the recovered file.
- Trusted forensic tools and repeatable procedures are used to reduce errors.
- Metadata, timestamps and content may be compared with other evidence to check consistency.
- Chain of custody is maintained so every transfer or access to the evidence is recorded.

## Sheet No. 19

**Q1. Create a table documenting recovered files, original path, recovery status, and evidential value.**

**Answer / Procedure:**

| Recovered File | Original Path                   | Recovery Status     | Evidential Value                        |
|----------------|---------------------------------|---------------------|-----------------------------------------|
| document.docx  | Example: /Users/User/Documents/ | Recovered           | May contain useful written information  |
| photo.jpg      | Example: /Users/User/Pictures/  | Recovered           | May provide visual or metadata evidence |
| notes.txt      | Example: /Users/User/Desktop/   | Partial / Recovered | May show user activity or intent        |

Use Autopsy or the given forensic tool to identify recovered items. For every file, note the original path, whether recovery is complete or partial, and why the file may be useful to the case. Replace the sample entries above with your actual findings.

**Result:** Recovered evidence was documented in a clear and structured table.

**Q2. Explain the importance of documentation in forensic recovery.**

**Answer:**

- Documentation creates a clear record of what the investigator found and what actions were performed.
- It records where a file was recovered from, how it was recovered and what tool was used.
- It helps maintain chain of custody and shows that the evidence was handled correctly.
- Good notes make the work repeatable so another examiner can review the same process.
- Clear documentation also makes it easier to prepare investigation reports and present reliable evidence.

## Sheet No. 20

**Q1. Perform deleted file recovery and prepare a concise forensic evidence summary suitable for an investigation report.**

**Answer / Procedure:**

- Open the forensic image in Autopsy and identify deleted files.
- Preview deleted files and select items that are relevant to the investigation.
- Recover the selected files to a separate evidence folder.
- Record important information such as original path, file type, size, timestamps and hash values.
- Briefly explain what was recovered and why it may be important.
- **Example evidence summary:** Deleted files were identified in the forensic image and recovered using Autopsy. Relevant file names, original locations, timestamps and hash values were recorded. The recovered items may help reconstruct user activity and support other evidence in the case.
- Keep the summary factual and avoid making conclusions that are not supported by the recovered data.
- **Result:** Deleted evidence was recovered and summarized in a form suitable for an investigation report.

**Q2. Explain evidence preservation after recovery.**

**Answer:**

- The original evidence should remain unchanged and should be stored securely.
- Recovered files should be saved in a separate evidence location with controlled access.
- Hash values should be calculated and recorded to detect any future change.
- Backups may be created, but each copy should be identified and handled properly.
- Chain-of-custody records must show who accessed or transferred the evidence and when.

## Sheet No. 26

### Q1. Capture network packets using Wireshark and identify DNS queries and HTTP traffic.

#### Answer / Procedure:

- Open Wireshark and choose the active network interface, such as Ethernet or Wi-Fi.
- Click **Start Capture** and generate normal network activity.
- After enough packets are captured, click **Stop**.
- Apply the display filter **dns** to view DNS queries and responses.
- Check domain names, source IP, destination IP and query/response details.
- Apply the display filter **http** to view unencrypted HTTP traffic if such traffic exists in the capture.
- Examine request methods, hosts, URLs and response details where available.
- Save the capture as a PCAP/PCAPNG file and record important packet numbers.
- **Result:** Network traffic was captured successfully and DNS/HTTP traffic was identified where present.

### Q2. Explain the importance of network forensics.

#### Answer:

- Network forensics is the examination of network traffic to find evidence of communication and attacks.
- It can identify source and destination IP addresses, ports, protocols, domains and connection times.
- Investigators can use it to detect suspicious connections, malware communication and unusual data transfers.
- Packet data helps reconstruct what happened across a network during an incident.
- Network evidence is useful in incident response, intrusion investigation and cybercrime cases.

## Sheet No. 27

**Q1. Analyze a provided packet capture and identify communicating IP addresses and protocols.**

**Answer / Procedure:**

- Open the provided PCAP or PCAPNG file in Wireshark.
- Look at the **Source** and **Destination** columns to see which IP addresses are communicating.
- Use **Statistics > Endpoints** to list systems found in the capture.
- Use **Statistics > Conversations** to identify common communication pairs.
- Check the **Protocol** column for protocols such as TCP, UDP, DNS, HTTP, TLS or ICMP.
- Apply display filters to inspect important traffic in more detail.
- Record the main IP addresses, protocols, ports and any unusual communication.
- **Result:** The major communicating IP addresses and network protocols were identified from the packet capture.

**Q2. Explain packet analysis in network investigations.**

**Answer:**

- Packet analysis means examining captured network packets to understand network activity.
- Investigators study source and destination IP addresses, ports, protocols, timestamps and packet contents.
- Filters make it easier to focus on specific traffic such as DNS, HTTP or a particular IP address.
- Packet analysis may reveal suspicious connections, failed communication, scanning or data transfers.
- It helps investigators reconstruct a sequence of network events during an incident.

## Sheet No. 29

**Q1. Examine a suspicious email and identify indicators of a phishing attempt. Prepare a report.**

**Answer / Procedure:**

- Check the sender email address carefully for misspellings, unusual domains or a mismatch with the display name.
- Read the subject and message for urgency, threats, prizes or unusual payment/password requests.
- Inspect links carefully and compare the visible text with the real destination domain.
- Check for unexpected attachments, especially executable files, archives or documents asking to enable macros.
- Look for poor grammar, generic greetings or branding that looks copied.
- Review the email header where available to identify sending servers and authentication results.
- Record each suspicious indicator separately in the report.
- Do not click suspicious links or open unknown attachments on a normal system.
- **Result:** The suspicious email was examined and common phishing indicators were documented.

**Q2. Explain common phishing techniques used in cyber attacks.**

**Answer:**

- **Email phishing:** Attackers send fake emails to many users and try to steal credentials or money.
- **Spear phishing:** A targeted message is created for a specific person or organization.
- **Smishing:** Phishing messages are sent through SMS or messaging services.
- **Vishing:** Attackers use phone calls or voice messages to trick victims.
- **Fake login pages:** A link opens a page designed to look genuine and steal usernames and passwords.
- **Malicious attachments:** Files may contain malware, scripts or documents that try to compromise the system.
- **Impersonation:** The attacker pretends to be a bank, company, manager or trusted person.